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Display module and display terminal

Abstract

A display module and a display terminal are provided. The display module includes a display panel, a support layer, and a buffer layer, wherein the display panel includes a display portion, a terminal bonding portion, and a bent portion. The support layer is disposed between the display portion and the terminal bonding portion, the buffer layer is disposed on the support layer, the buffer layer includes a first portion and a second portion disposed on a side of the first portion, the second portion corresponds to the terminal bonding portion, and rigidity of the second portion is greater than rigidity of the first portion.

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Background/Summary

TECHNICAL FIELD

(1) The present application relates to a field of display technology, in particular to a display module and a display terminal.

BACKGROUND

(2) A width of a frame of a flexible display module of an OLED can be narrowed by pad bending, and a bending radius of a pad is one of factors affecting the width of the frame of the display module. The less the bending radius of the pad, the narrower the width of the frame of the display module.

(3) Wherein it is critical for product stability of the display module to keep the bending radius of the pad without deformation or with a small amount of deformation. However, the pad of flexible display module is generally subjected to a large stress, and the bending radius of the pad is easily deformed during a pad bending process and a subsequent production process, resulting in poor product stability of the display module.

TECHNICAL PROBLEMS

(4) A current flexible display module has a technical problem that the bending radius of the pad is easily deformed during a pad bending process and a subsequent production process, resulting in poor product stability of the display module.

SUMMARY

(5) The present application provides a display module and a display terminal to improve a technical problem of a current flexible display module that the bending radius of the pad is easily deformed during a pad bending process and a subsequent production process, resulting in poor product stability of the display module.

(6) To solve the above technical problem, the present application provides the following technical solutions:

(7) The present application provides a display module comprising: a display panel comprising a display portion, a terminal bonding portion, and a bent portion disposed between the display portion and the terminal bonding portion, wherein the bent portion is bent away from a light emitting direction of the display panel; a support layer disposed between the display portion and the terminal bonding portion; and a buffer layer disposed on the support layer, wherein the buffer layer comprises a first portion, and a second portion disposed on a side of the first portion, and the second portion corresponds to the terminal bonding portion; and wherein rigidity of the second portion is greater than rigidity of the first portion.

(8) In the display module of the present application, an orthographic projection of the terminal bonding portion on the buffer layer is defined in the second portion.

(9) In the display module of the present application, an orthographic projection of a boundary between the terminal bonding portion and the bent portion on the buffer layer is defined in the second portion.

(10) In the display module of the present application, the first portion is disposed separately from the second portion.

(11) In the display module of the present application, a distance between the first portion and the second portion is greater than or equal to a thickness of the first portion or a thickness of the second portion in the light emitting direction of the display panel.

(12) In the display module of the present application, the first portion and the second portion constitute an intermediate layer of the buffer layer.

(13) In the display module of the present application, the buffer layer further comprises a first adhesive layer and a second adhesive layer disposed on two sides of the intermediate layer, respectively.

(14) In the display module of the present application, a ratio of a thickness of the first adhesive layer and a thickness of the second adhesive layer to a thickness of the intermediate layer in the

light emitting direction of the display panel is between $\frac{1}{4}$ and 2.

(15) In the display module of the present application, an orthographic projection of the first adhesive layer and the second adhesive layer on the intermediate layer is defined in the intermediate layer.

(16) In the display module of the present application, a distance between an edge of the first adhesive layer and an edge of the intermediate layer, and a distance between an edge of the second adhesive layer and the edge of the intermediate layer are both less than or equal to a distance between the first portion and the second portion.

(17) In the display module of the present application, the display portion comprises a display area, and a non-display area disposed on both sides of the display area, and the bent portion is connected to the display portion in the non-display area.

(18) In the display module of the present application, an orthographic projection of the second portion on the display portion extends into the display area.

(19) In the display module of the present application, the support layer comprises a first support portion, a second support portion, and a hollow portion disposed between the first support portion and the second support portion.

(20) In the display module of the present application, an orthographic projection of the second portion on the support layer is defined separated from the hollow portion.

(21) In the display module of the present application, the display module comprises a reinforcing plate disposed on a side of the support layer close to the terminal bonding portion.

(22) In the display module of the present application, an orthographic projection of the reinforcing plate on the support layer is defined in the second portion.

(23) In the display module of the present application, the display module comprises a second backplane disposed between the reinforcing plate and the terminal bonding portion.

(24) In the display module of the present application, the terminal bonding portion is fixed to the second backplane.

(25) In the display module of the present application, an orthographic projection of a boundary between the terminal bonding portion and the bent portion on the second backplane is defined in the second backplane.

(26) The present application also provides a display terminal comprising a terminal body and the display module, wherein the display module and the terminal body are integrated.

BENEFICIAL EFFECTS

(27) In the present application, the buffer layer positioned on the support layer is provided to include a first portion and a second portion, the second portion corresponds to the terminal bonding portion, and the rigidity of the second portion is greater than the rigidity of the first portion. On the basis of the support layer, the second portion having greater rigidity can further enhance the structural strength of the laminated structure at the position where the terminal bonding portion is positioned so that the bending radius of the bent portion is kept undeformed or a small amount of deformation, and the product stability of the display module is improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to more clearly illustrate the technical solutions in the embodiments of the present application, the accompanying drawings required for use in the description of the embodiments will be briefly described below. It will be apparent that the accompanying drawings in the following description are merely some embodiments of the present application, and other drawings may be obtained from these drawings without creative effort by those skilled in the art.

(2) FIG. 1 is a schematic diagram of an overall structure of a display module according to the

present application.

(3) FIG. 2 is a schematic structural diagram of a buffer layer according to the present application.

(4) FIGS. 3 to 6 are schematic diagrams of a production process of a buffer layer according to the present application.

REFERENCE NUMERALS

(5) a display panel **100**, a display portion **110**, a display area **111**, a non-display area **112**, a bent portion **120**, an ultraviolet curable glue **121**, a terminal bonding portion **130**, a support layer **200**, a first support portion **210**, a second support portion **220**, a hollow portion **230**, a buffer layer **300**, a first portion **310**, a second portion **320**, a first adhesive layer **330**, a second adhesive layer **340**, a protective film layer **350**, a release film **360**, a temporary protective film **370**, a first backplane **410**, a second backplane **420**, a polarizer **500**, an optical adhesive layer **600**, a flexible cover layer **700**, a scratch resistant layer **800**, and a reinforcing plate **900**.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(6) The following describes the technical solutions of the embodiments of the present application in a clear and complete manner with reference to the accompanying drawings in the embodiments of the present application. It will be apparent that the described embodiments are only a part of the embodiments of the present application, but not all of the embodiments. Based on the embodiments of the present application, all other embodiments obtained by those skilled in the art without creative effort fall within the scope of the present application. Furthermore, it should be understood that the specific embodiments described herein are intended only to illustrate and explain the present application and are not intended to limit the present application. The terms “up” and “down” are used in the present application in the absence of a reverse description, generally referring to the upper and lower parts of the device in actual use or operation, in particular in the drawing direction; and terms “in” and “out” are for the profile of the device.

(7) A width of a frame of a flexible display module of an OLED can be narrowed by pad bending, and a bending radius of a pad is one of factors affecting the width of the frame of the display module. The less the bending radius of the pad, the narrower the width of the frame of the display module.

(8) Wherein it is critical for product stability of the display module to keep the bending radius of the pad without deformation or with a small amount of deformation. However, the pad of flexible display module is generally subjected to a large stress, and the bending radius of the pad is easily deformed during a pad bending process and a subsequent production process, resulting in poor product stability of the display module.

(9) Referring to FIGS. 1 to 6, a display module includes a display panel **100**, a support layer **200**, and a buffer layer **300**. The display panel **100** includes a display portion **110**, a terminal bonding portion **130**, and a bent portion **120** between the display portion **110** and the terminal bonding portion **130**. The bent portion **120** is bent away from a light emitting direction of the display panel **100**. The support layer **200** is disposed between the display portion **110** and the terminal bonding portion **130**. The buffer layer **300** is disposed on the support layer **200**. The buffer layer **300** includes a first portion **310**, and a second portion **320** on a side of the first portion **310**. The second portion **320** corresponds to the terminal bonding portion **130**. Wherein rigidity of the second portion **320** is greater than rigidity of the first portion **310**.

(10) In the present application, the buffer layer **300** positioned on the support layer **200** is provided to include a first portion **310** and a second portion **320**, the second portion **320** corresponds to the terminal bonding portion **130**, and the rigidity of the second portion **320** is greater than the rigidity of the first portion **310**. On the basis of the support layer **200**, the second portion **320** having greater rigidity can further enhance structural strength of a laminated structure at the position where the terminal bonding portion **130** is positioned, so that the bending radius of the bent portion **120** is kept undeformed or slightly deformed, and the product stability of the display module is improved.

(11) The technical solutions of this application will now be described with reference to specific

embodiments. It should be noted that the order of description of the following embodiments is not a limitation on the preferred order of the embodiments.

(12) In an embodiment, the display portion **110**, the bent portion **120**, and the terminal bonding portion **130** are integrally formed. The display portion **110** is a main body portion of the display panel **100** for displaying. The bent portion **120** is positioned at an end of the display portion **110** and is bent away from the light emitting direction of the display panel **100**. The bent portion **120** may be provided with an ultraviolet curable adhesive **121** on a side away from the display portion **110**. The terminal bonding portion **130** is connected to an end of the bent portion **120** away from the display portion **110**, and the terminal bonding portion **130** is disposed on a side of the display portion **110** away from the light emitting direction of the display portion **110** and parallel to the display portion **110**.

(13) In this embodiment, the terminal bonding portion may be connected to an electrical component such as a flexible circuit board, a drive chip (not shown), or the like.

(14) In the present embodiment, the display panel **100** may be a flexible display panel **100**, such as an OLED panel, a mini-LED panel, a micro-LED panel, or the like.

(15) In an embodiment, the support layer **200** may be a rigid support material, such as a metal or alloy plate/sheet such as a stainless-steel plate/sheet.

(16) In an embodiment, the first portion **310** may be made of a relatively soft cushioning material, such as a sponge, rubber, or other foamed material.

(17) In an embodiment, the second portion **320** may be made of a sheet material having a relatively great rigidity, such as a metal sheet, an alloy sheet, a stainless-steel sheet, or the like.

(18) Referring to FIG. **1**, in the display module of the present application, the display module may further include a first backplane **410** disposed on the buffer layer **300**, a polarizer **500** disposed on the display portion **110**, an optical adhesive layer **600** disposed on the polarizer **500**, a flexible cover layer **700** disposed on the optical adhesive layer **600**, and a scratch resistant layer **800** disposed on the flexible cover layer **700**.

(19) In an embodiment, a display portion **110** of the display panel **100** is disposed on the first backplane **410**, and an orthographic projection of the display portion **110** on the first backplane **410** is in the first backplane **410**.

(20) In an embodiment, the first backplane **410** may be made of a flexible organic material such as polyimide (PI), polyester (BOPET), cycloolefin polymer, liquid crystal polymer (LCP), polydimethylsiloxane (PDMS), or the like.

(21) In an embodiment, the polarizer **500** may be composed of a polyvinyl alcohol material layer (PVA), a cellulose triacetate material layer (TAC), a pressure sensitive adhesive layer, a release film **360**, and a protective film layer **350**.

(22) In an embodiment, the optical adhesive layer **600** may be made of materials such as silicone rubber, acrylic resin, unsaturated polyester, polyurethane, and epoxy resin.

(23) In an embodiment, the flexible cover layer **700** may be an ultra-thin glass (UTG) cover, a transparent polyimide (PI) cover.

(24) In an embodiment, the scratch resistant layer **800** may be made of an organic film layer material having good scratch resistance, such as polyethylene terephthalate (PET) material or the like.

(25) Referring to FIG. **1**, in the display module of the present application, an orthographic projection of the terminal bonding portion **130** on the buffer layer **300** is in the second portion **320**, so that all of the buffer layer **300** corresponding to the terminal bonding portion **130** is made of a material having greater rigidity, thereby improving stiffness and strength of a position where the terminal bonding portion **130** connects with other film layer of the display module, so that the bent portion **120** connected to the terminal bonding portion **130** is not easily deformed, so that the bending radius of the bent portion **120** is kept without deformation or slightly deformed, thereby improving product stability.

(26) Since the boundary between the terminal bonding portion **130** and the bent portion **120** is subjected to a greater bending stress, the terminal bonding portion **130** is easily peeled from or displaced from another film layers of the display module at this position, resulting in a change in the bending radius of the bent portion **120**. Therefore, in the present embodiment, an orthographic projection of the boundary between the terminal bonding portion **130** and the bent portion **120** on the buffer layer **300** is in the second portion **320**.

(27) In an embodiment, the boundary between the terminal bonding portion **130** and the bent portion **120** is provided at a position opposite to the second portion **320**, so that the second portion **320** can provide a good structural strengthen effect on the boundary position between the terminal bonding portion **130** and the bent portion **120**, and improve structural strength of the terminal bonding portion **130** bonded to another film layers of the display module.

(28) Referring to FIG. 1 and FIG. 2, FIG. 2 is a schematic structural diagram of the buffer layer according to the present application. In the display module of the present application, the first portion **310** and the second portion **320** are separated from each other, so as to avoid or reduce mechanical interference due to different materials when the first portion **310** and the second portion **320** are directly connected, thereby ensuring normal display.

(29) In an embodiment, a distance $d1$ between the first portion **310** and the second portion **320** is greater than or equal to a thickness $d2$ of the first portion **310** or the second portion **320** in the light emitting direction of the display panel **100**. In an embodiment, the first portion **310** and the second portion **320** are provided so as to always maintain a sufficient distance with respect to the specifications of the display module, thereby better avoiding interference between the first portion **310** and the second portion **320** and ensuring display quality.

(30) In an embodiment, according to the size of the display module, the distance $d1$ between the first portion **310** and the second portion **320** may be any value within 0.05 mm to 0.3 mm depending on the specification and size of the display module. In an embodiment, the distance $d1$ between the first portion **310** and the second portion **320** is not more than 0.3 mm, so that the distance $d1$ between the first portion **310** and the second portion **320** is not too large to form a film bonding step difference, and further, a problem of imprinting due to the step difference is avoided.

(31) In an embodiment, in the light emitting direction of the display panel **100**, the thickness of the first portion **310** may be equal to the thickness of the second portion **320**, so that surfaces of the first portion **310** and the second portion **320** are flush, and the imprint caused by the step difference is further reduced. In an embodiment, the thickness $d2$ of the first portion **310** and the thickness $d2$ of the second portion **320** in the light emitting direction of the display panel **100** may be from 50 μm to 100 μm .

(32) Referring to FIG. 2, in the display module of the present application, the first portion **310** and the second portion **320** constitute an intermediate layer of the buffer layer **300**, and the buffer layer **300** further includes a first adhesive layer **330** and a second adhesive layer **340** disposed on a side of the intermediate layer, respectively. In an embodiment, the first adhesive layer **330** and the second adhesive layer **340** are disposed on either side of the intermediate layer formed by the first portion **310** and the second portion **320**, so that the first adhesive layer **330** and the second adhesive layer **340** can fix and protect the first portion **310** and the second portion **320**, and the first adhesive layer **330** and the second adhesive layer **340** can be bonded to other film layers of the display module.

(33) In an embodiment, the first adhesive layer **330** and the second adhesive layer **340** may be made of a same material as the optical adhesive layer **600**.

(34) In an embodiment, referring to FIGS. 3 to 6, the buffer layer **300** may be separately stored by being covered with a protective film layer **350** on a surface before being combined with another film layers of the display module, that is, the protective film layer **350** may cover a side surface of the first adhesive layer **330** and the second adhesive layer **340** away from the intermediate layer. The protective film layer **350** on the surface is removed when used.

(35) In an embodiment, the buffer layer **300** may be made by following steps: **S100**: cutting the first adhesive layer **330** and the first portion **310** of the desired size according to the specification.

(36) In an embodiment, referring to FIG. 3, the first adhesive layer **330** may be attached to the protective film layer **350**, and the first adhesive layer **330** may be coated with the release film **360** on a surface of the first adhesive layer **330** away from the protective film layer **350** after cutting. **S200**: bonding the first portion **310** to the side surface of the first adhesive layer **330**.

(37) In an embodiment, referring to FIG. 4, a part of the release film **360** on the first adhesive layer **330** may be removed first, then the first portion **310** may be bonded at a position where the release film **360** is removed, and then the temporary protective film **370** may be coated on the surface of the first portion **310** again. **S300**: cutting the second portion **320** of the desired size according to the specification. **S400**: bonding the second portion **320** to the first adhesive layer **330**, leaving a gap between the second portion **320** and the first portion **310**.

(38) In an embodiment, referring to in FIG. 5, a remaining part of the release film **360** on the first adhesive layer **330** may be removed first, then the second portion **320** is bonded to the remaining portion of the release film **360**, and then the temporary protective film **370** is coated on the surfaces of the first portion **310** and the second portion **320**, respectively. **S500**: cutting the second adhesive layer **340** of the desired size according to the specification. **S600**: bonding the second adhesive layer **340** to the surfaces of the first portion **310** and the second portion **320** away from the first adhesive layer **330**.

(39) In an embodiment, referring to FIG. 6, the temporary protective film **370** on the surfaces of the first portion **310** and the second portion **320** is removed first, then the second adhesive layer **340** is bonded to the surface, and then another protective film layer **350** is bonded to the second adhesive layer **340**.

(40) Referring to FIG. 2, in an embodiment, a ratio of thicknesses $d3$ of the first adhesive layer **330** and the second adhesive layer **340** to a thickness of the intermediate layer (i.e., the thicknesses $d2$ of the first portion **310** and the second portion **320**) in the light emitting direction of the display panel **100** is $\frac{1}{4}$ to 2. Specifically, as described above, in the light emitting direction of the display panel **100**, the thicknesses $d2$ of the first portion **310** and the second portion **320** may be 50 μm to 100 μm , and the thickness $d3$ of the first adhesive layer **330** and the second adhesive layer **340** may be 25 μm to 100 μm .

(41) In an embodiment, the thicknesses $d3$ of the first adhesive layer **330** and the second adhesive layer **340** are set to be 25 μm to 100 μm , so that the first adhesive layer **330** and the second adhesive layer **340** have good fixing and protection effects, and the laminated thickness of the display module can be prevented from being increased due to excessive thickness, thereby facilitating lightening of the display module.

(42) Referring to FIG. 1, in the display module of the present application, an orthographic projection of the first adhesive layer **330** and the second adhesive layer **340** on the intermediate layer is in the intermediate layer to ensure that edges of the first adhesive layer **330** and the second adhesive layer **340** do not extend beyond the intermediate layer, thereby reducing a risk of glue overflow.

(43) In an embodiment, a distance $d4$ between the edge of the first adhesive layer **330** and the edge of the intermediate layer, and between the edge of the second adhesive layer **340** and the edge of the intermediate layer is smaller than or equal to the distance $d1$ between the first portion **310** and the second portion **320**. Specifically, as described above, the distance $d1$ between the first portion **310** and the second portion **320** may be 0.05 mm to 0.3 mm, and the distance $d4$ between the edge of the first adhesive layer **330** and the edge of the intermediate layer, and between the edge of the second adhesive layer **340** and the edge of the intermediate layer may be 0 to 0.02 mm.

(44) In an embodiment, the distance $d4$ between the edge of the first adhesive layer **330** and the edge of the intermediate layer, and between the edge of the second adhesive layer **340** and the edge of the intermediate layer is set to be 0 to 0.02 mm, so that the risk of glue overflow can be

effectively reduced, and the first adhesive layer **330** and the second adhesive layer **340** can sufficiently cover the first portion **310** and the second portion **320**, thereby achieving better fixing and protection effects.

(45) Referring to FIG. **1**, in the display module of the present application, the display portion **110** includes a display area **111** and a non-display area **112** disposed on both sides of the display area **111**, the bent portion **120** is connected to the display portion **110** in the non-display area **112**, and an orthographic projection of the second portion **320** on the display portion **110** extends into the display area **111**.

(46) In an embodiment, the second portion **320** can extend below the display area **111** of the display portion **110**, thereby increasing an area ratio of the second portion **320** in the intermediate layer, and further improving structure enhancement effect of the second portion **320** on the position of the terminal bonding portion **130**.

(47) Referring to FIG. **1**, in the display module of the present application, the support layer **200** may include the first support portion **210**, the second support portion **220**, and the hollow portion **230** between the first support portion **210** and the second support portion **220**. The hollow portion **230** corresponds to the display area **111** of the display portion **110**. When the flexible display module is used to dynamically fold a display product, the flexible display module may be bent at the position of the hollow portion **230**.

(48) In an embodiment, an orthographic projection of the second portion **320** on the support layer **200** is provided separately from the hollow portion **230**, that is, the second portion **320** may extend into the display area **111** of the display portion **110**, but the second portion **320** cannot exceed a position corresponding to the hollow portion **230** so that the orthographic projection of the second portion **320** in the support layer **200** does not overlap the hollow portion **230**. In the present embodiment, the second portion **320** having greater rigidity does not interfere with the bending of the display module in the hollow portion **230**, thereby ensuring the dynamic bending performance of the display module.

(49) Referring to FIG. **1**, in a display module of the present application, the display module further includes a reinforcing plate **900** disposed on a side of the support layer **200** close to the terminal bonding portion **130**, and an orthographic projection of the reinforcing plate **900** on the support layer **200** is in the second portion **320**.

(50) In an embodiment, the reinforcing plate **900** may be made of a polyimide (PI) material, a polyethylene terephthalate (PET) material, a copper-clad plate material, an aluminum sheet, a steel sheet, or the like.

(51) In an embodiment, the reinforcing plate **900** is provided so that an orthographic projection of the reinforcing plate **900** on the support layer **200** is in the second portion **320**, thereby enabling the second portion **320** to support and strengthen all positions of the reinforcing plate **900**, thereby improving the reinforcing effect of the reinforcing plate **900**.

(52) Referring to FIG. **1**, in the display module of the present application, the display module may further include a second backplane **420** disposed between the reinforcing plate **900** and the terminal bonding portion **130**, and the terminal bonding portion **130** is fixed to the second backplane **420**.

(53) In an embodiment, the second backplane **420** may be made of a same material as the first backplane **410**, i.e., the first backplane **410** and the second backplane **420** may be made of a flexible organic material such as polyimide (PI), polyester (BOPET), cycloolefin polymer, liquid crystal polymer (LCP), polydimethylsiloxane (PDMS), or the like, as described above.

(54) In an embodiment, an orthographic projection of the boundary between the terminal bonding portion **130** and the bent portion **120** on the second backplane **420** is in the second backplane **420**, so that the second backplane **420** can provide good support for the boundary position between the terminal bonding portion **130** and the bent portion **120**, effectively alleviate the stress concentration problem at the boundary position, further improve bending radius stability of the bent portion **120**, and further improve the product stability of the display module.

(55) An embodiment of the present application further provides a display terminal comprising a terminal body and the display module, wherein the display module and the terminal body are integrated. In an embodiment, the display terminal may be an intelligent electronic display device such as a mobile phone, a computer, a television, or a wristwatch.

(56) The display module and display terminal provided in the embodiments of the present application are described in detail above. The principles and embodiments of the present application are described in detail herein. The description of the embodiments is merely intended to help understand the method and core ideas of the present application. At the same time, a person skilled in the art may make changes in the specific embodiments and application scope according to the idea of the present application. In conclusion, the content of the specification should not be construed as a limitation to the present application.

Claims

1. A display module, comprising: a display panel comprising a display portion, a terminal bonding portion, and a bent portion disposed between the display portion and the terminal bonding portion, wherein the bent portion is bent away from a light emitting direction of the display panel; a support layer disposed between the display portion and the terminal bonding portion; and a buffer layer disposed on the support layer, wherein the buffer layer comprises a first portion and a second portion disposed on a side of the first portion, and the second portion is closer to the terminal bonding portion than the first portion; wherein rigidity of the second portion is greater than rigidity of the first portion; the first portion and the second portion are linearly spaced apart, and materials of the second portion and the support layer are rigid materials, and a material of the first portion is a soft cushioning material.
2. The display module according to claim 1, wherein an orthographic projection of the terminal bonding portion on the buffer layer is defined in the second portion.
3. The display module according to claim 2, wherein an orthographic projection of a boundary between the terminal bonding portion and the bent portion on the buffer layer is defined in the second portion.
4. The display module according to claim 1, wherein a distance between the first portion and the second portion is greater than or equal to a thickness of the first portion or a thickness of the second portion in the light emitting direction of the display panel.
5. The display module according to claim 1, wherein the first portion and the second portion constitute an intermediate layer of the buffer layer.
6. The display module according to claim 5, wherein the buffer layer further comprises a first adhesive layer and a second adhesive layer disposed on two sides of the intermediate layer, respectively.
7. The display module according to claim 6, wherein a ratio of a thickness of the first adhesive layer and a thickness of the second adhesive layer to a thickness of the intermediate layer in the light emitting direction of the display panel is between $\frac{1}{4}$ and 2.
8. The display module according to claim 6, wherein an orthographic projection of the first adhesive layer and the second adhesive layer on the intermediate layer is defined in the intermediate layer.
9. The display module according to claim 8, wherein a distance between an edge of the first adhesive layer and an edge of the intermediate layer, and a distance between an edge of the second adhesive layer and the edge of the intermediate layer are less than or equal to a distance between the first portion and the second portion.
10. The display module according to claim 3, wherein the display portion comprises a display area and a non-display area disposed on both sides of the display area, and the bent portion is connected to the display portion in the non-display area.

11. The display module according to claim 10, wherein an orthographic projection of the second portion on the display portion extends into the display area.
 12. The display module according to claim 1, wherein the display module comprises a reinforcing plate disposed on a side of the support layer close to the terminal bonding portion.
 13. The display module according to claim 12, wherein an orthographic projection of the reinforcing plate on the support layer is defined in the second portion.
 14. The display module according to claim 13, wherein the display module comprises a second backplane disposed between the reinforcing plate and the terminal bonding portion.
 15. The display module according to claim 14, wherein the terminal bonding portion is fixed to the second backplane.
 16. The display module according to claim 15, wherein an orthographic projection of a boundary between the terminal bonding portion and the bent portion on the second backplane is defined in the second backplane.
 17. A display terminal, comprising a display module, comprising: a display panel comprising a display portion, a terminal bonding portion, and a bent portion disposed between the display portion and the terminal bonding portion, wherein the bent portion is bent away from a light emitting direction of the display panel; a support layer disposed between the display portion and the terminal bonding portion; and a buffer layer disposed on the support layer, wherein the buffer layer comprises a first portion and a second portion disposed on a side of the first portion, and the second portion is closer to the terminal bonding portion than the first portion; wherein rigidity of the second portion is greater than rigidity of the first portion; the first portion and the second portion are linearly spaced apart, and materials of the second portion and the support layer are rigid materials, and a material of the first portion is a soft cushioning material.
 18. The display module according to claim 11, wherein the support layer comprises a first support portion, a second support portion, and a hollow portion disposed between the first support portion and the second support portion.
 19. The display module according to claim 18, wherein an orthographic projection of the second portion on the support layer is defined separated from the hollow portion.
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